

RESECTION: ABSTRACT INTRODUCTION

Name/Group: _____

Day: _____

Purpose

Resection finds an unknown position using two or more visible landmarks whose positions are known. Measure the azimuth from you to each landmark, calculate each back azimuth, and draw the back-azimuth lines from the landmarks. Your position is where those lines intersect.

Essential ideas

Azimuth: an angle measured clockwise from north, from 000 through 360 degrees.

Line of sight: the straight line between an observer and a visible landmark.

Back azimuth: the same line viewed in the opposite direction.

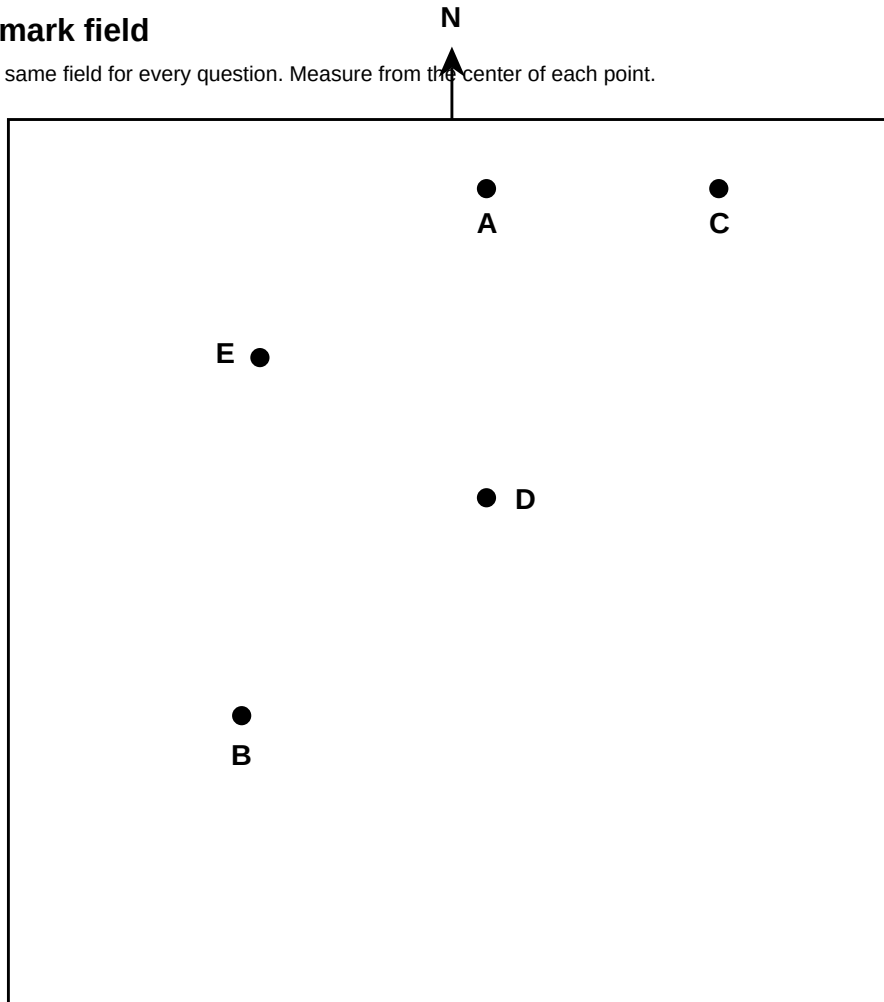
Rule: below 180, add 180. At or above 180, subtract 180.

Direction - azimuth reference

N 000	NNE 022.5	NE 045	ENE 067.5	E 090	ESE 112.5	SE 135	SSE 157.5
S 180	SSW 202.5	SW 225	WSW 247.5	W 270	WNW 292.5	NW 315	NNW 337.5

Landmark field

Use this same field for every question. Measure from the center of each point.



DIRECTION AND AZIMUTH PRACTICE

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Use page 1. Name the closest cardinal, intercardinal, or secondary-intercardinal direction, then measure the specific azimuth to the nearest degree. Do not infer the number from the name.

Measure both forms

1. D from A: Direction _____ Azimuth _____
2. A from D: Direction _____ Azimuth _____
3. D from B: Direction _____ Azimuth _____
4. B from D: Direction _____ Azimuth _____
5. E from D: Direction _____ Azimuth _____
6. D from E: Direction _____ Azimuth _____
7. C from A: Direction _____ Azimuth _____
8. A from C: Direction _____ Azimuth _____
9. A from B: Direction _____ Azimuth _____
10. E from C: Direction _____ Azimuth _____

Back azimuths - show +180 or -180

- | | | |
|------------------|-----------------|-----------------|
| 1. 050 -> _____ | 2. 300 -> _____ | 3. 022 -> _____ |
| 4. 151 -> _____ | 5. 182 -> _____ | 6. 000 -> _____ |
| 7. 090 -> _____ | 8. 247 -> _____ | 9. 330 -> _____ |
| 10. 360 -> _____ | | |

Explain the relationship

1. Compare B-to-D with D-to-B. What mathematical relationship do you notice?

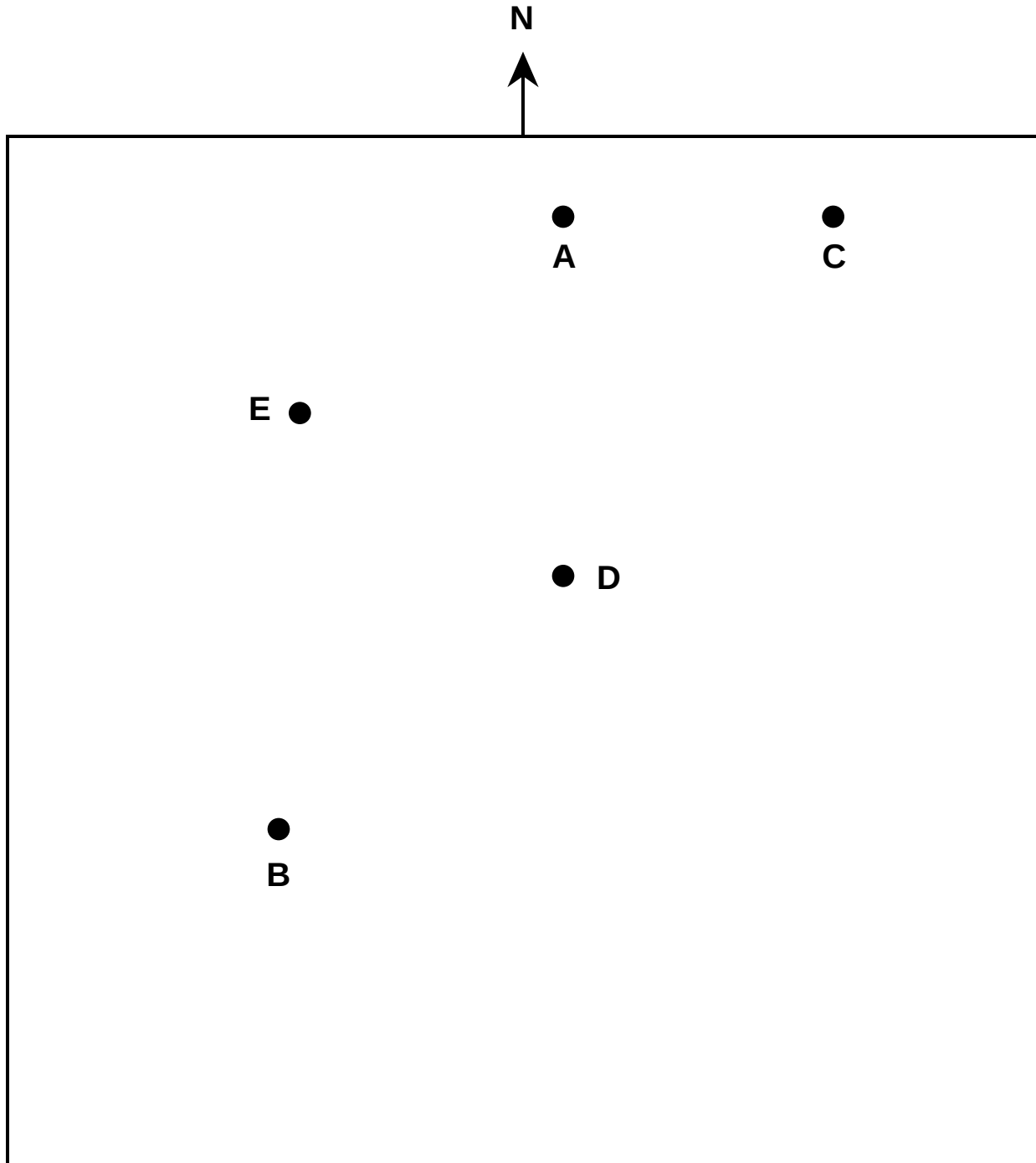
2. Why is the back azimuth drawn from a landmark during resection?

LOCATING UNKNOWN POSITIONS

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Plot each position on the clean field. Use a small circled number matching the question.
Draw only the lines needed. Unless told halfway or center, intersect two lines of sight.



1. You are halfway between E and A. Plot and label position 1.
2. You are at the center of the triangle formed by E, B, and D. Plot and label position 2.
3. You are due south of C and due east of D. Plot and label position 3.

INTRODUCTORY RESECTION CHALLENGES

Name/Group: _____

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Bearings are measured from your unknown position TO visible landmarks. Convert them to back azimuths before drawing from the landmarks. Plot solutions on page 3 and label each number.

Two-landmark fixes

4. A at 018 deg; E at 293 deg. Back azimuths: A _____ E _____

5. C at 035 deg; D at 228 deg. Back azimuths: C _____ D _____

6. B at 230 deg; E at 300 deg. Back azimuths: B _____ E _____

Direction-language fixes

7. Landmark A is northwest of you. Landmark C is northeast of you.

8. Landmark E is WNW of you. Landmark D is SSE of you.

9. Landmark B is SSW of you. Landmark D is ENE of you.

Mixed fixes

10. Landmark A is due north of you. Your measured azimuth to B is 230 deg.

11. Landmark E is northwest of you. Your measured azimuth to C is 045 deg.

Reflection

1. Why do two lines of sight usually locate you better than one?

2. How does a small measurement error affect lines that cross at a shallow angle?

3. Which landmark pair gives the strongest fix from the center, and why?
